

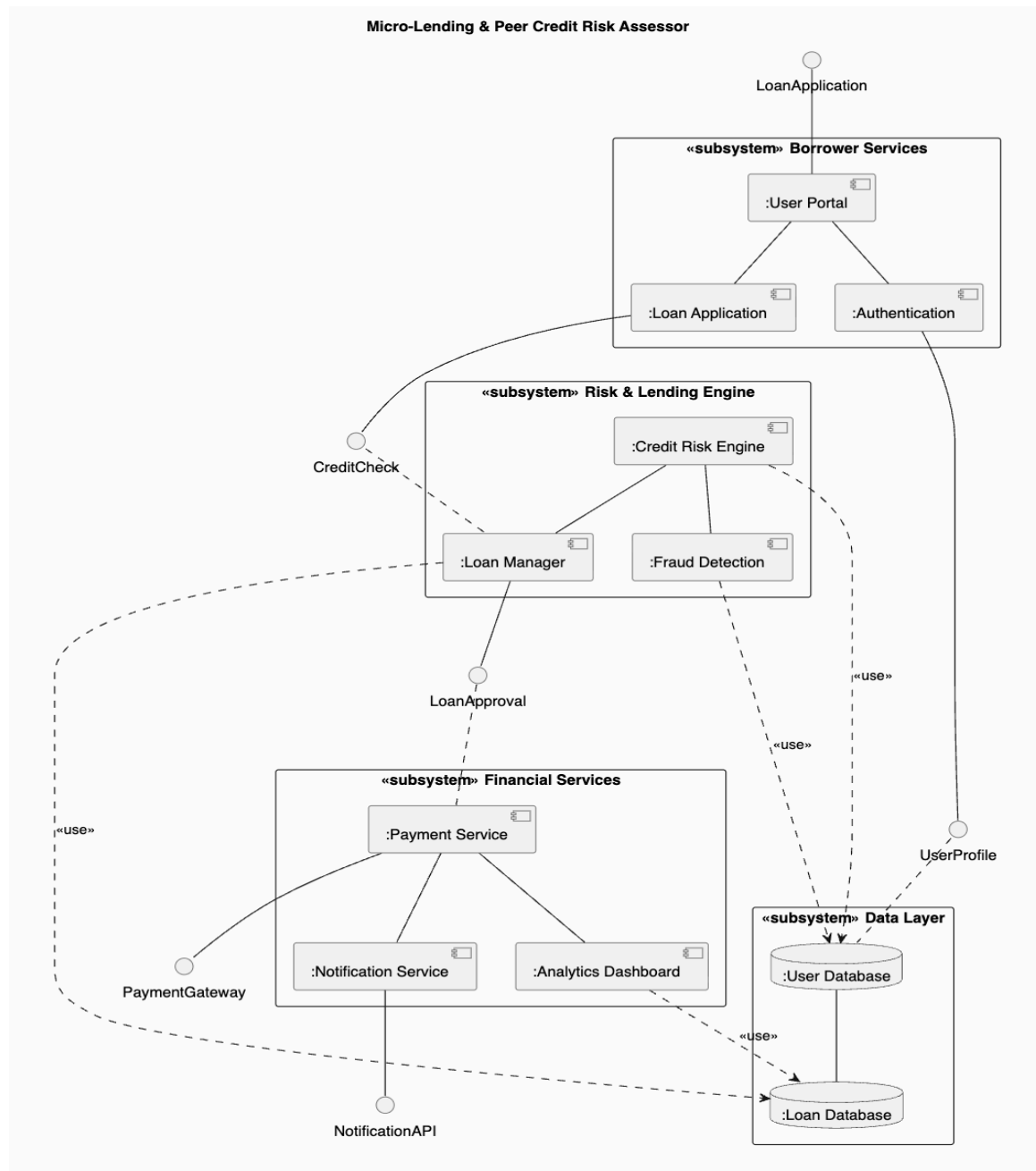
SE LAB 3

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TOPIC: Micro-Lending & Peer Credit Risk Assessor

DIAGRAM:



Micro-Lending & Peer Credit Risk Assessor

Component Modelling & Architecture Report

1. Project Overview

The Micro-Lending & Peer Credit Risk Assessor is a software platform designed to facilitate secure peer-to-peer lending by connecting borrowers and lenders while assessing borrower creditworthiness using a dedicated risk assessment engine. The system automates loan applications, credit evaluation, payment processing, notifications, and analytics through modular software components.

2. Selected Architecture

Chosen Architecture: Layered Architecture

The system is organized into four logical subsystems:

- Borrower Services (Presentation Layer)
- Risk & Lending Engine (Business Layer)
- Financial Services (Service Layer)
- Data Layer (Persistence Layer)

This separation ensures that each subsystem performs a specific responsibility while communicating through well-defined interfaces.

3. Architectural Justification

Reason 1: Clear Separation of Concerns

Each subsystem handles a distinct responsibility. User-facing operations remain isolated from business logic, while payment processing and database management operate independently, making the system easier to maintain and extend.

Reason 2: Improved Maintainability

Features such as fraud detection, analytics, or payment gateways can be updated independently without affecting the entire application, reducing maintenance effort and improving scalability.

4. Component Description

The component diagram contains multiple interacting components grouped into four subsystems.

Subsystem	Components	Responsibility
Borrower Services	User Portal, Loan Application, Authentication	User interaction and authentication
Risk & Lending Engine	Credit Risk Engine, Loan Manager, Fraud Detection	Loan evaluation and risk analysis
Financial Services	Payment Service, Notification Service, Analytics Dashboard	Payments, notifications, reporting
Data Layer	User Database, Loan Database	Data storage

5. Component Interactions

The system communicates through six major interfaces:

- LoanApplication – receives borrower loan requests.
- CreditCheck – sends applications to the Credit Risk Engine.
- LoanApproval – forwards approved loans to the Payment Service.
- PaymentGateway – processes financial transactions.
- NotificationAPI – sends approval and repayment notifications.
- UserProfile – retrieves authenticated user information.

